

Data Pipeline Part 2

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Volume of classes hours: 25 hrs ($\pm$ same personal work expected)

Course summary:

This course covers the fundamentals of data engineering, focusing on Apache Spark, Kafka, and modern data platform components in particular data lakehouses and the iceberg table format. It focuses on the “engineering” part of “data engineering”. In the course we will build an ETL application using pyspark on AWS Glue. We will deploy the infrastructure using Terraform. We will test+deploy the code using Gitlab CI/CD. We will orchestrate it with Airflow.

Course Objectives:

The course aims to provide students with a solid understanding of data engineering concepts and practical skills in designing and building data pipelines. By the end of the course, students should be able to build and deploy a batch processing ETL application using the iceberg table format in AWS.

Theoretical background used:

The course draws on concepts from distributed computing, data structures, algorithms, and software engineering. It covers topics such as stream processing, query optimization, data modeling, containerization, and DevOps practices.

Technologies used:

The course covers Apache Spark, AWS, Terraform, and modern CI/CD tools such as Gitlab.

Course Pre-Requisites:

- Data Pipeline Part 1
- Big Data Ecosystem by Adaltas
- DevOps
- Software Engineering
- Amazon AWS “Cloud-Computing”
- Machine Learning with Python Labs

Course evaluation:

MCQ (100%)

There will be an MCQ in the end of the course.

According to DSTI Scientific Advisory Board policy for ever-evolving programmes, this syllabus may be subject to adaptations and changes when the class will be delivered by the selected Professor(s).